Project details

Log-Log Legends · DTU 02805 · state on 14 November 2026 · the one-page version is *overview.pdf*. Each analysis gets a short block: what it asks, where it lives, what it found.

Data and shared rules

What	Count	Source
Movies with 10,000+ votes (the graph)	12,673	IMDb
Clean films with budget and gross in USD	2,446 (100k+), 6,851 (10k+)	Wikipedia infoboxes
Screenplays	1,095 clean 100k+ films	MovieSum
Audience reviews	2,070 clean 100k+ films, to 2017	MovieLens Tag Genome
Critic scores	Metacritic 2,431, RT 2,740	Wikipedia, Wikidata

"Clean" means budget and gross both in USD and at least $1,000, no rentals-only or regional gross, no streaming distributor, and released before 2026. The 100k+ tier favours hits: only 169 of its 2,446 films grossed less than their budget. Profit is ln(gross/budget) everywhere. Network measures use only earlier films, every result gets a null or a second method, and no script text, review text or IMDb fields go into git.

Who made the film

Movie graph (code, notebook)

Movies, people (by role and billing) and companies in one network, with studio names merged through Wikidata IDs.

- The studio's track record is the strongest single measure: $\rho = +0.27$ with the multiple, net of budget and year, on 4,568 films.
- All network measures together lift R^2 from 0.041 to 0.130 over budget, year and genre (shuffle $p = 0.005$).
- A director's next film is a hit 35% of the time after a hit, and 18% after a non-hit.

Shared-people returns (code)

Do films with the same director, producer, cast member or studio have more similar returns than chance?

- Every role clusters: 0.09 (top-three cast) to 0.17 (producers), against a shuffle null of at most 0.02.
- It survives dropping franchise films, ranking the returns, and keeping only films from 1980 on. About half survives a shuffle within each studio.
- It's descriptive: it uses all years at once.

Actor network (Niklas, branch branch, not merged)

Top-three billed actors, linked by shared films. It comes with an interactive page with projections, communities, centrality and per-actor ROI.

What the film says

Scripts and reviews notebook (notebook)

- Only 2 of 15 script features pass $q < 0.05$ net of budget and year. The strongest is a character network centred on one character ($\rho = -0.097$).
- Six VADER arc shapes don't differ in profit ($p = 0.99$).

Scattertext words (code, page, pre-registered)

Top against bottom quarter by multiple, within genre and decade, on dialogue.

- The difference is slightly above random groupings ($z = 2.08$, $p = 0.024$).
- On the franchise check's own null (200 shuffles), it drops from $z = 2.05$ with all films to $z = 0.50$ without franchise films. Dropping the same number of random films never does that (0 of 50 tries), so the excess goes with the franchise films themselves.
- A word classifier doesn't beat genre, decade and budget (AUC 0.58 against 0.58).

Emotional arcs (code, page, pre-registered, one test)

Every sentence of 834 non-franchise scripts is scored by DistilBERT SST-2. Each film becomes a 100-point tone curve, summarised as swing (D1) and ending lift (D2). Budget, genre, decade and scene length are controlled.

- Swing: no link ($\rho = +0.035$, $p = 0.32$).
- Ending lift: a near miss ($\rho = +0.078$, $p = 0.028$, against a pre-set cutoff of 0.025). Its sign holds in 12 of 13

variants, but not with VADER.
- The arc adds nothing to out-of-sample prediction.
- About half of the arc variation between films is scene noise (split-half reliability around 0.45).
- Exploratory replication of Del Vecchio et al. (2021): their "Man in a hole" effect does not reproduce on our scripts, even in their own specification (code).
- A fourth sentiment model, ModernBERT, is running as a robustness check.

How it was received

Reviews exploration (Àngela, branch, not merged)

2,095 films up to 2017. Every correlation is tested against a shuffle within genre × decade.

- The multiple goes with IMDb rating at $\rho = 0.27$, Metacritic 0.26 and Rotten Tomatoes 0.33. Genre and decade alone give about 0.1.
- The critics agree with each other far more (Metacritic and Rotten Tomatoes at 0.93). Review tone goes with the multiple at 0.13.
- A disagreement score (rating rank minus money rank) finds "liked beyond their money" films (*tarantino*, *noir*) and "earned beyond their liking" films (*scares*, *slasher*).

What predicts profit (notebook)

Trained on films to 2010, tested on 2011 to 2017.

Model	Out-of-sample R^2
Budget, year, genre	-0.010
+ network	0.202
All pre-release blocks (adds month, franchise)	0.244
+ critic and audience scores	0.312

- On films with a screenplay, script features add nothing beyond the network (R^2 0.037 against 0.036).
- Director hit streaks beat a shuffle of each director's own films (+0.175 against +0.090). PageRank adds nothing beyond plain experience.
- Rotten Tomatoes goes with the multiple at $\rho = +0.25$. There are 158 loved flops and 205 panned hits.

Open and weak spots

- **Human validation**, which only people can fill:
  - Scattertext sheets
  - arc scene sheets and the extreme-arc sheet
  - the 50-review sample on the reviews branch
- **Arcs Step 11** waits on the instrument question (whole-review VADER against sentence-level VADER).
- **Weak spots**: the franchise flag is rough, release month isn't checked, reviews stop at 2017, and gross over budget isn't profit.

Where things are

Everything links to the group repository on GitHub.

- Repository and its README (data, how to get it, how to run each part).
- Step-by-step notebooks: emotional arcs, word comparison, movie graph, profit model, scripts and reviews.
- Every decision and test, dated: arcs log, word-comparison log.
- Pre-registrations: arcs config, word-comparison config.
- Run guides: arcs, word comparison.
- The paper sources (also in Overleaf as "Log-Log Legends - 02805 paper (PNAS)").